

Data-Driven Stock Analysis using Time Series Models

1. Methodology and Models Used

This project bridges the gap between theoretical time series analysis and live market execution. The methodology relies on fetching historical daily data via Yahoo Finance, preprocessing it for stationarity, and utilizing two distinct quantitative models to forecast short-term price action: **Facebook Prophet** (an additive regression model handling seasonality) and **Auto-ARIMA** (a classical autoregressive integrated moving average model). These forecasts, integrated with historical volatility metrics, dictated a strictly quantitative portfolio allocation strategy that was subsequently executed and validated on the live StockGro platform.

2. Stock Selection Rationale (Task 1)

To ensure optimal portfolio diversification and mitigate sector-specific unsystematic risk, a universe of 5 highly liquid large-cap stocks was selected across distinct key sectors of the Indian economy:

- **RELIANCE (Energy/Conglomerate):** Acts as a high-weight market anchor with a steady long-term growth trend.
- **HDFCBANK (Banking/Financials):** Selected from a high-momentum sector to capture core economic cycles.
- **INFY (Information Technology):** Provides exposure to global export demand and exhibits distinct seasonal patterns.
- **TMCV (Automobile):** The newly demerged Tata Motors Commercial Vehicle listing, chosen for its unique volatility profile and high momentum.
- **ITC (FMCG):** Acts as a defensive, low-beta asset to stabilize the portfolio during broader market swings.

3. Data Preprocessing Steps (Task 2)

To optimize the models for real-world execution on May 14 and May 15, 2026, the dataset timeframe was strictly defined up to the close of May 13, 2026. The following preprocessing steps were applied:

- **Handling Missing Values & Demerger Artifacts:** Standard forward fill (`ffill`) was applied to account for market holidays. Crucially, a backward fill (`bfill`) was implemented to account for the TMCV demerger artifact, successfully preventing the deletion of 4 years of historical data across the rest of the portfolio.
- **Stationarity Testing:** The Augmented Dickey-Fuller (ADF) test was utilized to check for unit roots. Non-stationary series were differenced internally by the ARIMA framework to achieve stationarity.
- **Validation Split:** The data was dynamically split, isolating the final 6 months (Nov 13, 2025 – May 13, 2026) exclusively for out-of-sample model validation prior to the final live-forecasting run.

4. Forecast Results with Confidence Intervals (Task 3)

The models successfully projected the expected closing prices for the live 2-day trading window utilizing the full preprocessed dataset.

Confidence Intervals: The Prophet model natively generated 95% confidence bounds (`yhat_lower` and `yhat_upper`) for each prediction, mathematically capturing the expected variance and potential market drift. The table below represents the final ensemble average derived from both the Prophet and Auto-ARIMA baselines:

Stock	Day 1 Forecast (May 14)	Day 2 Forecast (May 15)
RELIANCE	₹1,411.08	₹1,411.20
HDFCBANK	₹792.94	₹793.81
INFY	₹1,145.13	₹1,144.41
TMCV	₹414.89	₹415.34
ITC	₹306.47	₹305.90

5. Volatility and Trend Analysis Findings (Task 4)

Short-term risk was quantified using the rolling standard deviation of logarithmic returns. Structural trend behavior was isolated using STL (Seasonal and Trend decomposition using Loess) over a 30-day window. *(Note: TMCV's volatility registered artificially low due to the necessary bfill applied during Task 2, rendering it a data artifact).*

Stock	Daily Volatility (σ)	Annualized Risk	Core Trend Phase
RELIANCE	0.0144	22.92%	Sideways/Consolidating
HDFCBANK	0.0137	21.68%	Sideways/Consolidating
INFY	0.0155	24.68%	Sideways/Consolidating
TMCV	0.0082*	12.97%*	Sideways/Consolidating
ITC	0.0129	20.54%	Sideways/Consolidating

6. Portfolio Composition and Allocation Rationale (Task 5)

A hybrid allocation approach was deployed, leveraging both mathematical strategies and human-in-the-loop qualitative overrides for a ₹10,00,000 portfolio:

- **RELIANCE (25% / ₹2,50,000):** Forecast-Guided (Strategy A). Core portfolio anchor due to steady underlying baseline trend and high market liquidity.
- **HDFCBANK (22% / ₹2,20,000):** Forecast-Guided (Strategy A). Strong momentum capture with high intraday stability.
- **ITC (20% / ₹2,00,000):** Volatility-Aware (Strategy B). Selected as the safest *true* asset (20.54% Annualized Risk) based on historical market realities to serve as the primary defensive stabilizer.
- **INFY (18% / ₹1,80,000):** Forecast-Guided (Strategy A). Moderate sizing based on positive upside, but capped at 18% because Task 4 correctly identified it as the highest-risk historical asset (24.68%).
- **TMCV (15% / ₹1,50,000):** Human-in-the-Loop Override. Task 4 mathematically calculated TMCV's Annualized Risk at an artificially low 12.97%. Recognizing this as a backward-fill

data artifact of a newly listed entity, the lowest portfolio weight was assigned to protect capital from unquantified volatility.

7. Model Comparison and Accuracy Evaluation (Task 6)

Both models were evaluated on the final 6-month out-of-sample test split.

- **Auto-ARIMA:** Excelled at capturing recent, short-term momentum shifts via its moving average errors, but struggled slightly with sudden non-linear market shocks.
- **Facebook Prophet:** Exhibited superior handling of missing data and strong overarching seasonal components (e.g., quarterly IT earnings cycles), and provided vital confidence intervals. However, it occasionally smoothed out volatile spikes too aggressively.
- **Conclusion:** An ensemble approach combining the structural stability of Prophet with the short-term responsiveness of ARIMA provided the most robust baseline forecasting model.

8. StockGro Execution Summary (Task 7)

The calculated weights were strictly executed on the StockGro platform on **Day 1 (May 14)** prior to the market close.

Stock	Quantity Bought	Execution Price	Total Capital Deployed
HDFCBANK	291	₹742.11	₹2,15,954.01
ITC	651	₹299.84	₹1,95,195.84
INFY	162	₹1,095.73	₹1,77,508.26
TMCV	439	₹370.40	₹1,62,603.49
RELIANCE	183	₹1,340.47	₹2,45,306.01

Total Capital Deployed: ₹9,96,567.61

9. Predicted vs Actual Outcomes (Task 8)

At the market close of **Day 2 (May 15)**, the final realized portfolio performance was isolated and assessed against the initial model forecasts.

Realized Portfolio Returns

- **HDFCBANK:** ₹762.33 (+2.72% / +₹5,884.02)
- **ITC:** ₹305.84 (+2.00% / +₹3,906.00)
- **INFY:** ₹1,106.18 (+0.95% / +₹1,692.90)
- **TMCV:** ₹372.90 (+0.67% / +₹1,099.61)
- **RELIANCE:** ₹1,322.53 (-1.34% / -₹3,283.02)
- **Final Net Portfolio Return: +0.93% (+₹9,299.51)**

Absolute Percentage Error Evaluation

The following table compares the ensemble model forecasts for Day 2 against the live executed closing prices.

Stock	Predicted Price (Day 2)	Actual Live Price	Absolute % Error
ITC	₹305.90	₹305.84	0.02%
INFY	₹1144.41	₹1,106.18	3.46%
HDFCBANK	₹793.81	₹762.33	4.13%
RELIANCE	₹1411.20	₹1,322.53	6.70%
TMCV	₹415.34	₹372.90	11.38%

10. Reflections

The strategy successfully translated theoretical forecasting into a profitable live-market execution, generating a +0.93% return over the strict 2-day window.

1. **What Worked (Volatility Sizing Success):** Overriding the algorithm and manually assigning ITC a heavy defensive 20% weight while limiting TMCV to 15% was highly effective. ITC, correctly identified as the true lowest-volatility asset, forecasted with a near-perfect 0.02% margin of error, serving as a major profit driver while minimizing downside risk.
2. **What Didn't (Algorithmic Lag):** The live execution highlighted the inherent limitations of purely autoregressive time-series models. The models naturally overshot the prices of high-momentum stocks like Reliance and TMCV because they failed to account for overnight market gap-downs and unsystematic real-world shocks that occurred leading into the May 14 open.
3. **What Could Be Improved:** Future iterations of this algorithmic pipeline must include sentiment analysis (NLP) on financial news feeds to predict sudden gap-downs. Furthermore, implementing dynamic weighting filters to flag and exclude artificially smoothed data (like the TMCV demerger artifact) would further automate risk mitigation.